

KIRSTEN ZIMAN

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ACADEMIC APPOINTMENTS

Princeton University Lecturer Princeton Neuroscience Institute	<i>1/2025 - Present</i>
University of Pennsylvania Lecturer College of Liberal and Professional Studies	<i>6/2025 - 8/2025</i>
Princeton University Postdoctoral Associate Princeton Neuroscience Institute Advisor: Michael Graziano	<i>2024 - 2025</i>
Princeton University Postdoctoral Fellow Princeton Neuroscience Institute Advisor: Michael Graziano	<i>2022 - 2024</i>

EDUCATION

Dartmouth College Ph.D., Cognitive Neuroscience Psychological and Brain Sciences Advisor: Jeremy Manning	<i>2022</i>
University of Southern California B.S., Neuroscience Brain and Creativity Institute Advisors: Antonio Damasio, Assal Habibi	<i>2014</i>

FUNDING & ACCOLADES

Data Driven Social Science Grant, Princeton University <i>Principal Investigator, Aim: Modeling attention to social scenes in healthy controls vs. patients with schizophrenia</i>	<i>2026</i>
Data Driven Social Science Grant, Princeton University <i>Principal Investigator, Aim: Developing introspective AI using attentional neural networks</i>	<i>2025</i>

NIH Institutional Training Grant (T32MH065214)
Postdoctoral Research Fellow

2022-2024

NSF Established Program to Stimulate Competitive Research (Award #1632738) *Graduate Student* 2017-2022

Graduate Travel Award, Dartmouth Graduate Studies 2021
Neukom Prize for Outstanding Graduate Research in Computational Science 2020
Tau Sigma Honors Society, University of Southern California 2013
Phi Theta Kappa Honors Society, Beta Kappa Delta Chapter 2012

PREPRINTS & MANUSCRIPTS IN PREPARATION

Undergraduate trainees are underlined

2. Pieslinger J., **Ziman K.**, Igelstrom K., Graziano M. S. A. (In preparation). Judging the attention of others is robust to variation in social cognition.

1. Farrell K. T., **Ziman K.**, Graziano M. S. A. (Under Review). Improving how agents cooperate: attention schemas in artificial neural networks. *Preprint: <https://arxiv.org/abs/2411.00983>*

PEER-REVIEWED PUBLICATIONS

Undergraduate trainees are underlined

9. Premakumar V. N., Vaiana M., Pop F., Rosenblatt J., Schwerz de Lucena, D., **Ziman K.**, Graziano M. S. A. (2026). Unexpected benefits of self-modeling in neural systems. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 384(2320).

8. **Ziman K.**, Kimmel S. C., Christian I., Farrell K. T., Graziano M. S. A. (2025). Cortical networks active when modeling the attention of others. *Cerebral Cortex*, 35(9), bhaf266.

7. Christian I., Samuel A. Nastase, Mindy Yu, **Ziman K.**, Graziano M. S. A. (2025). Monitoring attentional state in self and others. *Journal of Cognitive Neuroscience*, 1-11.

6. **Ziman K.**, Kimmel S. C., Farrell K. T., Graziano M. S. A. (2023). Predicting the attention of others. *Proceedings of the National Academy of Sciences*, 120(42), e2307584120.

5. Saleki, S., **Ziman, K.**, Hartstein, K. C., Cavanagh, P., Peter, U. T. (2022). Endogenous attention biases transformational apparent motion based on high-level shape representations. *Journal of Vision*, 22(12), 16-16.

4. Hartstein K. C., Saleki S., **Ziman K.**, Cavanagh P., Tse P. U. (2021). First- and second-order transformational apparent motion rely on common shape representations. *Vision Research*, 188,

3. **Ziman K.**, Heusser A. C., Fitzpatrick P. C., Field C. E., & Manning J. R. (2018). Is automatic speech-to-text transcription ready for use in psychological experiments? *Behavior Research Methods*, 50(6), 2597-2605.

2. Heusser A. C.*, **Ziman K.***, Owen L. L. W., & Manning J. R. (2018). HyperTools: A Python toolbox for gaining geometric insights into high-dimensional data. *Journal of Machine Learning Research*. 18(152), 1-6. * Indicates equal first authorship

1. Heusser A. C., Fitzpatrick C. P., Field C. E., **Ziman K.**, & Manning J. R. (2017). Quail: A Python toolbox for analyzing and plotting free recall data. *The Journal of Open Source Software*, 2(18).

TALKS

Modeling the Attention of Others, *Rutgers-Princeton Center for Computational Cognitive NeuroPsychiatry*, 3/26/2025, Virtual.

Predicting the Attention of Others, *Barnard Vision Lab*, 9/25/2024, New York, NY.

Attention Modeling and Social Cognition. *Princeton Neuroscience Institute Seminar Series*, 3/21/2024, Princeton, NJ.

Attentional modeling in the Brain. *Society for Neuroscience*, 11/11/2023, Washington, DC.

Modeling and Predicting the Attention of Others, *Association for the Scientific Study of Consciousness*, 6/22/2023, New York, NY.

Pupil dilation increases when participants report familiarity for images of faces they have not seen before, *Association for the Scientific Study of Consciousness*, 6/16/2021, Virtual.

First & Second-order Transformational Apparent Motion Rely on Common Shape Representations, *Barnard Vision Lab*, 11/15/2021, Virtual.

Cognitive Markers of Psychiatric Traits, *Dartmouth College Cognitive Brown Bag Series*, 2020, Hanover, NH.

Attention and Memory, *Dartmouth College Specialist Presentation*, 2019, Hanover, NH.

Volitional Attention Modulates Encoding and Retrieval, *Dartmouth College Cognitive Brown Bag Series*, 2018, Hanover, NH.

Speaker at EPSCoR Attention Consortium Talk Series, *Montana State University*, 2017, Bozeman, Montana.

POSTERS & ABSTRACTS

Undergraduate trainees are underlined

Ziman K., Thompson J. (2025). Towards introspective AI. *Princeton Neuroscience Institute Retreat*. Asbury Park, NJ.

Kimmel S. C., **Ziman K.**, Christian I., Thompson J., Graziano M. S. A. (2024). Predictive modeling of others' attention in the brain: Increased activity when expectations are violated. *Society for Neuroscience*. Chicago, IL.

Farrell K. T., **Ziman K.**, Graziano M. S. A. (2024). The attention schema theory in machine learning: training agents to classify the attention patterns of others. *Society for Neuroscience*. Chicago, IL.

Ziman K., Kimmel S. C., Farrell K. T., Graziano M. S. A. (2024). Neural activity when predicting the attention of others. *Princeton Neuroscience Institute Retreat*. Atlantic City, NJ.

Kimmel S. C., Farrell K. T., **Ziman K.**, Graziano M. S. A. (2023). Modeling and Predicting the Attention of Others. *Association for the Scientific Study of Consciousness*. New York, NY.

Ziman K., Kimmel S. C., Farrell K. T., Graziano M. S. A. (2023). Predicting the Attention of Others. *Princeton Neuroscience Institute Retreat*. Philadelphia, PA.

Ziman K., Manning J. R. (2021). Increased pupil dilations are associated with unexpected false familiarity for faces. *Society for Neuroscience*. Chicago, IL.

Ziman K., Manning J. R. (2021). Pupil dilation increases when participants report familiarity for images of faces they have not seen before. *Association for the Scientific Study of Consciousness*. Virtual from Tel-Aviv, Israel.

Ziman K., Lee M. R., Martinez A. R., Manning J. R. (2019). Volitional attention modulates memory encoding and retrieval. *Society for Neuroscience Conference*. Chicago, IL.

Ziman K., Heusser A. C., Fitzpatrick P. C., Field C. E., & Manning J. R. (2018). Is automatic speech-to-text transcription ready for use in psychological experiments? *Context and Episodic Memory Symposium*. Philadelphia, PA.

Fitzpatrick P. C., **Ziman K.**, Heusser A. C., Field C. E., Manning J. R. (2018) The utility of speech-to-text software for transcription of verbal response data. *Wetterhan Science Symposium*. Hanover, NH.

Pak E.K., **Ziman K.**, Manning JR (2018) How does attention affect memory? *Wetterhan Science Symposium*. Hanover, NH.

Lee M. R., Chacko RS, Whitaker EC, Fitzpatrick P. C., Field C. E., **Ziman K.**, Bollinger BJ, Heusser AC, Manning JR (2018) Adaptive Free Recall: Enhancing (Or Diminishing) Memory. *Wetterhan Science Symposium*. Hanover, NH.

Ziman K., Heusser A.C., Manning J.R. (2017). Effects of Study Context on Recall Organization. *Society for Neuroscience Conference*. Washington, DC.

Ziman K., Heusser A.C., Manning J.R. (2017). Harnessing the power of mnemonic fingerprints: Maximizing learning potential by personalizing stimulus organization during adaptive list learning. *Context and Episodic Memory Symposium*. Philadelphia, PA.

Heusser A.C., **Ziman, K.**, Manning J.R. (2017). HyperTools: A Python toolbox for visualizing and manipulating high-dimensional data. *Context and Episodic Memory Symposium*. Philadelphia, PA.

Manning JR, **Ziman, K.**, Heusser AC (2017) Efficient Learning: Manipulating context to enhance (or diminish) memory. *Society for Neuroscience*. Washington, DC.

Ziman, K., Familiar, A.M., Shim, W.M. (2016). Positive affect worsens ensemble coding performance. *Vision Science Society Conference*. Saint Pete Beach, Florida.

Deirdre B., Lennon J., **Ziman K.** (2016). Content of Sleep-talking Transcripts versus Dream Accounts and Waking Language. *International Association for the Study of Dreams Conference*. Virginia Beach, Florida.

Wong, W.O., Suthana, N.A., Pourshaban, D., **Ziman, K.**, Bookheimer, S., Fried, I., Knowlton, B. (2012). Comparison of Medial Temporal Subregional Thickness and Overall Brain Volume to Episodic Memory Performance in Humans. *UCLA Undergraduate Poster Session*. Los Angeles, California.

SOFTWARE DEVELOPMENT

HyperTools toolbox for analyzing & visualizing high-dimensional data (Python, open-source)
*featured in Kaggle “No Free Hunch” blog and over 1,500 stars on GitHub

AutoFR toolbox to automatically transcribe verbal free recall data (Python, open-source)

Quail toolbox for analyzing and plotting free recall data (Python, open-source)

AD HOC REVIEWER

Nature Communications
Journal of Open Source Software
Journal of Experimental Psychology: General
Frontiers Open Access Research Journal
Cognition

TEACHING

Computational Models of Cognition, Princeton University *Fall 2026*
Preceptor with Professor Brenden Lake

Functional Neuroanatomy, Princeton University *Fall 2026*
Head Preceptor with Professor Michael Graziano

From Molecules to Systems to Behavior (Graduate), Princeton University *Spring 2026*
Lecturer with Professor Jonathan Cohen

Introduction to Cognitive Neuroscience, Princeton University *Spring 2026*
Head Preceptor with Professor Jesse Gomez

Functional Neuroanatomy, Princeton University *Fall 2025*
Head Preceptor with Professor Michael Graziano

Introduction to Experimental Psychology, University of Pennsylvania *Summer 2025*
Lecturer

Introduction to Cognitive Neuroscience, Princeton University *Spring 2025*
Head Preceptor with Professor Jesse Gomez

From Molecules to Systems to Behavior (Graduate), Princeton University Preceptor with Professor Jonathan Cohen	<i>Spring 2025</i>
Quantitative Methods , Princeton University Preceptor with Lecturer Justin Junge	<i>Spring 2025</i>
Functional Neuroanatomy , Princeton University Preceptor with Professor Michael Graziano	<i>Fall 2024</i>
Introduction to Programming in Python , Princeton University Course designer and instructor; six-week undergraduate workshop	<i>Summers 2024-2026</i>
Laboratory in Principles of Neuroscience , Princeton University Preceptor with Lecturer Anthony Ambrosini	<i>Spring 2023</i>
Functional Neuroanatomy , Princeton University Preceptor with Professor Michael Graziano	<i>Fall 2022</i>
Laboratory in Psychological Science , Dartmouth College Teaching Assistant to Professor Keilah Worth	<i>Winter 2020</i>
Experiment Design, Methodology & Data Analysis , Dartmouth College Teaching Assistant to Professor Catherine Cramer	<i>Spring 2019</i>
Principles of Human Brain Mapping with MRI , Dartmouth College Teaching Assistant to Professor Jeremy Huckins	<i>Fall 2019</i>
Principles of Human Brain Mapping with MRI , Dartmouth College Teaching Assistant to Professor Jeremy Huckins	<i>Fall 2018</i>

MENTORSHIP

GRADUATE MENTEES:

1. Yeo Bi Choi, *graduate student in the Robertson Laboratory at Dartmouth College*
2. Byeol Kim, *graduate student in the Wager Laboratory at Dartmouth College*

UNDERGRADUATE MENTEES:

1. Jack Thompson, *recipient of ReMatch+ research funding, Princeton University*
2. Sarah Kimmel, *recipient of Lenfest Scholar Foundation Special Education Expenses Fund*
3. Paxton Fitzpatrick, *current graduate student at Dartmouth College*
4. Ethan Adner, *recipient of Neukom undergraduate research fellowship at Dartmouth College*
5. Natalie Schroeder, *recipient of David C. Hodgson Endowment at Dartmouth College*
6. Darren Gu, *recipient of David C. Hodgson Endowment at Dartmouth College*
7. Madeline Lee, *Sophomore Research Scholar at Dartmouth College*

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| 8. Kathryn Farrell | 12. Alex Chivers | 16. Eowyn Pak |
| 9. Campbell Field | 13. Marisol Tracy | 17. Alejandro Martinez |
| 10. Sarah Park | 14. William Chen | 18. Chetan Pavuluri |
| 11. Chelsea Uddenberg | 15. Swestha Jain | 19. Christina Lu |

PROFESSIONAL ACTIVITIES

Association for the Scientific Study of Consciousness Student Committee Chair, 2023-2024
 Princeton Neuroscience Institute Social Committee, 2023
 Association for Women in Science member, 2022
 Association for the Scientific Study of Consciousness Student Committee Member, 2022
 National Center for Faculty Development & Diversity Member, 2022
 National Postdoctoral Association Member, 2022
 Interviewer for Dartmouth College graduate recruitment, 2021
 Dartmouth College representative at Society for Neuroscience, 2021
 Leader of Graduate Student Roundtable weekly meetings (one academic year), 2018
 Methods in Neuroscience at Dartmouth Workshop Attendee, 2017
 Leader of Attention Consortium (EPSCoR) Graduate Student Journal Club, 2017

EDUCATIONAL OUTREACH

The Science of Meditation and Yoga guided meditation for undergraduate students, 12/12/24
 HomeWork residential student visit to Princeton Neuroscience Institute, 2/5/24
 Re-Match+ undergraduate mentorship program at Princeton University, 2023-2024
 Scientific Reviewer for Mass STEM Hub, 2023 (reviewing high school research proposals)
 Mill Hill student laboratory tour and science activity, 2023
 Association for the Scientific Study of Consciousness Panelist, Career Panel, 2023
 Letters to a Pre-Scientist STEM professional letter writer, 2020
 Guest Speaker at Brain Speaker Series, Richmond Middle School 2017, 2019-2020
 Presenter at Dartmouth College Brain Bee, 2018

EMPLOYMENT HISTORY

Contextual Dynamics Laboratory, Dartmouth College	<i>2016-2017</i>
Laboratory Manager	
Perception and Cognition Laboratory, Dartmouth College	<i>2015-2016</i>
Laboratory Manager	
Brain and Creativity Institute, University of Southern California	<i>2013-2014</i>
Research Assistant	
Cognitive Neurophysiology Lab, University of California, Los Angeles	<i>2011-2012</i>
Research Assistant	

CLINICAL EXPERIENCE

Children's Hospital Los Angeles, Medical Preceptorship Program	<i>2013</i>
UCLA Orthopedic Hospital, Child Life Volunteer	<i>2011</i>